



PROFESSIONAL CERTIFICATION

Grasshopper Training

The Grasshopper Course teaches parametric design using visual programming in Rhino. Learners explore algorithmic modeling geometry manipulation and data structures. Through practical projects participants create complex forms optimize designs and generate variations. This training equips architects and designers with tools to innovate in computational design enhancing creativity



35-40

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

Grasshopper Course

It is the tool to learn how to use Grasshopper for design in Rhino.

Course Description

This course delves into Grasshopper for parametric architecture and design. Master visual scripting create generative geometries and integrate with Rhino for advanced modeling. Hands-on exercises cover data management optimization and fabrication preparation enabling innovative solutions in architecture product design and urban planning.

Duration

35-40

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Master Grasshopper interface.
- ✓ Create parametric models.
- ✓ Manipulate geometries.
- ✓ Manage data structures.
- ✓ Optimize design variations.
- ✓ Integrate with Rhino.
- ✓ Enhance creative processes.

Target Audience

- Designers seeking parametric tools.
- Draftsmen for 3D modeling.
- 3D Modellers in architecture.

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic Rhino knowledge required.
- Basic computer skills needed.
- Interest in algorithmic design.

Grasshopper Course

Start with Grasshopper fundamentals and parametric modeling.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
35 Hrs

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
38 Hrs

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
40 Hrs

Grasshopper Course

Start with Grasshopper fundamentals and parametric modeling.

01 Exploring Grasshopper Interface and Rhino Integration

- > Configure canvas layout and component library access
- > Establish live links between Rhino and Grasshopper geometry
- > Navigate component categories and quick-search functionality
- > Customize interface colors fonts and preview settings
- > Diagnose common connection and error display messages

02 Creating Basic Parametric Points and Curves

- > Generate points using coordinates and number parameters
- > Construct lines polylines and interpolated curves
- > Create circles ellipses and arcs parametrically
- > Build spiral and helix curves with variable controls
- > Position curve start and end points dynamically

03 Building Simple Surfaces and Basic Solids

- > Extrude curves into surfaces with height control
- > Loft multiple section curves into smooth surfaces
- > Revolve profiles around axes to form solids
- > Sweep profiles along paths with orientation options
- > Create ruled surfaces between two boundary curves

04 Handling Data Lists and Simple Operations

- > Create and manage single-level data lists
- > Perform basic arithmetic on list elements
- > Sort reverse and shift list item positions
- > Cull pattern and remove duplicate items
- > Display list contents using panel components

05 Applying Geometric Transformations Parametrically

- > Translate geometry using vector inputs
- > Rotate objects around custom axes and angles
- > Scale geometry uniformly or directionally
- > Mirror objects across planes or lines
- > Orient instances toward target points or vectors

06 Controlling Parameters with Sliders and Graphs

- > Create number sliders with defined domains
- > Use graph mapper for nonlinear value distribution
- > Remap numbers across custom input ranges
- > Link multiple sliders for coordinated control
- > Animate parameters to preview design behavior

Grasshopper Course

Start with Grasshopper fundamentals and parametric modeling.

07 Constructing and Managing Data Tree Structures

- > Graft and flatten lists into tree structures
- > Create and manage data tree branches
- > Shift paths and simplify tree hierarchy
- > Extract specific items from tree branches
- > Match data trees across multiple streams

08 Generating Complex Curves and Surfaces

- > Create NURBS curves with control point editing
- > Build surfaces from points or curve networks
- > Blend adjacent surfaces maintaining continuity
- > Divide surfaces into isocurve patterns
- > Extract edges borders and surface frames

09 Performing Boolean Operations on Solids

- > Union multiple closed solids together
- > Subtract one solid from another volume
- > Intersect solids to create new geometry
- > Split breps using cutting planes or surfaces
- > Resolve common boolean failure conditions

10 Analyzing Geometry Properties and Performance

- > Measure curve lengths and surface areas
- > Calculate volumes and centroid locations
- > Evaluate surface curvature and continuity
- > Generate contour sections at variable intervals
- > Display principal stress directions on surfaces

11 Creating Repeating Patterns and Modular Arrays

- > Generate linear and grid-based element arrays
- > Distribute objects in radial circular patterns
- > Create voronoi cells from point distributions
- > Apply attractor-based scaling and rotation
- > Build honeycomb and lattice structure systems

12 Implementing Geometric and Parametric Constraints

- > Enforce minimum and maximum distance rules
- > Maintain equal lengths or angles between elements
- > Prevent collision between moving components
- > Align objects to reference geometry automatically
- > Preserve planarity in complex surface networks

Grasshopper Course

Start with Grasshopper fundamentals and parametric modeling.

13 Building Reusable Custom Components and Clusters

- > Group logic into encapsulated cluster definitions
- > Create user objects with named inputs/outputs
- > Document custom components with descriptions
- > Share and import reusable definition libraries
- > Optimize cluster performance and readability

14 Running Physics Simulations and Form-Finding

- > Set up particle-spring structural systems
- > Apply forces supports and anchor conditions
- > Achieve equilibrium using dynamic relaxation
- > Simulate hanging-chain and minimal surface forms
- > Evaluate structural behavior under load variations

15 Optimizing Designs with Evolutionary Algorithms

- > Define clear multi-objective fitness functions
- > Run genetic algorithms on parametric variables
- > Analyze trade-offs using Pareto front visualization
- > Combine simulation results with optimization loops
- > Select and export optimal design solutions

16 Integrating AI and Machine Learning Workflows

- > Connect Grasshopper to external AI prediction models
- > Generate design variants using trained networks
- > Optimize parameters through learned relationships
- > Evaluate AI-suggested solutions against constraints
- > Automate iterative refinement with AI feedback

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

Up to 30%

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Free

Demo Session



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